Year 1 AS Statistics and Mechanics

Chapter 3

Representations of data

Practice Booklet

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3.1 Outliers

Practice Book, printed page 17.

1. The scores achieved by a group of students on a maths test are recorded.
The quartiles are: $Q_1 = 50\%$ and $Q_3 = 65\%$.
A value greater than $Q_3 + 1.5 \times (Q_3 - Q_1)$ or smaller than $Q_1 - 1.5 \times (Q_3 - Q_1)$ is defined as an outlier.
Work out whether the following scores are outliers, according to this definition:

a. 12%

b. 71%

c. 90%

Hint

Use the definition of an outlier supplied in the question, and work out the lower and upper bounds.

2. Some data is collected. The mean and standard deviation are calculated and found to be 14.5 and 6.2 respectively.
An outlier is defined as an observation which lies more than 2 standard deviations from the mean.
Work out whether the following are outliers using this rule:

a. 12.5

b. 27.1

c. 26.9

Hint

There are many possible definitions of an outlier. You will be told which definition to use in the question.

3. E/P The lengths of a colony of rattlesnakes are recorded. The mean length is 1.5 m and the variance is 0.04.
An outlier is an observation which lies ±2 standard deviations from the mean.
The length of one of the rattlesnakes is recorded as 2.1 m.

a. State, with a reason, whether this is an outlier. (2 marks)

b. Calculate the largest and smallest lengths that a rattlesnake can be without being considered an outlier. (2 marks)

c. State, with a reason, whether this recorded length should be removed from the data. (2 marks)

4. E/P The ages of 20 people on a coach trip are recorded. The lower quartile of the ages is found to be 22 and the upper quartile is found to be 46.
An outlier is an observation that falls either 1.5× the interquartile range below the lower quartile, or 1.5× the interquartile range above the upper quartile.

a. Calculate the minimum age and maximum age a person on the trip could be without being considered an outlier. (3 marks)

One of the people on the trip has an age recorded as 103.

b. Suggest a reason why this could be a legitimate data value. (1 mark)

5. E/P The ages of 10 students at a secondary school are recorded.
The summary statistics are $\sum x = 154$ and $\sum x^2 = 2438$.

a. Calculate the mean and standard deviation of the ages. (3 marks)

An outlier is an observation which lies ± 2 standard deviations from the mean.
One of the ages recorded is 21.

b. State, with a reason, whether this is an outlier. (2 marks)

Poppy decides to exclude this data value from her analysis.

c. Give one reason to justify this decision. (1 mark)

d. Calculate the mean and standard deviation of the 9 remaining students. (3 marks)

6. E/P The daily mean temperatures, °C, are recorded for a random sample of 12 days in Beijing.

18.5 26.1 27.4 27.9 28.1 28.2 29.4 30.9 30.4 27.2 30.0 42.6

a. Calculate the mean and standard deviation of the daily mean temperatures. (2 marks)

An outlier is an observation which is ± 2 standard deviations from the mean.

b. Show that there are two outliers in the data and state their values. (2 marks)

c. Suggest, with reasons, whether the outliers should be excluded from the data. (2 marks)

d. Clean the data and recalculate the mean and standard deviation. (2 marks)

3.2 Box plots

Practice Book, printed page 18.

1. A group of learner drivers took a test in which they had to identify road hazards.
Their scores are summarised as follows:

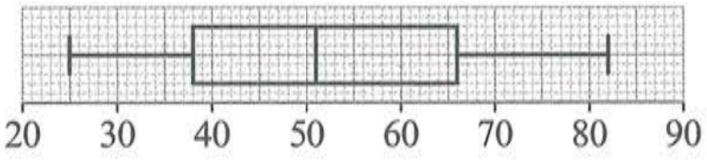
Lower quartile = 30	Median = 42	Upper quartile = 45
Lowest score = 18	Highest score = 52	

Given that there are no outliers, draw a box plot to illustrate this data.

Hint

Draw little vertical lines at each key value from the summary data.

2. The diagram shows a box plot of the goals scored by football teams over a season.



Write down:

a. the median

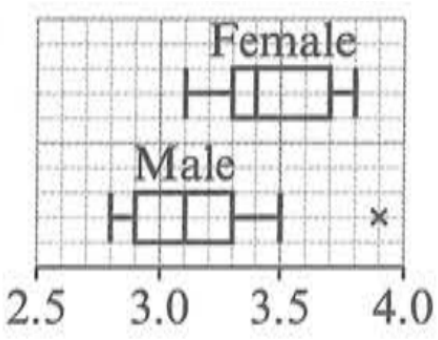
b. the interquartile range

c. the range.

Hint

Read the quartiles and maximum and minimum values from the box plot.

3. E/P The masses, in grams, of a group of male and female ruby-throated hummingbirds are recorded. The data is summarised in two box plots.



a. Explain what is represented by × in the box plot for the male hummingbirds. (1 mark)

b. Compare the masses of the male and female hummingbirds. (2 marks)

One of the hummingbirds is found to have a mass of 3.6 g.

c. State, with a reason, whether this hummingbird is likely to be male or female. (1 mark)

4. E/P Data for the daily mean windspeed (in knots) in Leuchars is July 1987 is taken from the large data set.

3	4	5	5	5	5	5	5	5	5	6
6	6	7	7	7	8	8	8	8	9	9
9	9	10	11	11	12	15	16	19		

a. Calculate the median and the interquartile range. (3 marks)

An outlier is defined as a value which lies either 1.5× the interquartile range above the upper quartile or 1.5× the interquartile range below the lower quartile.

b. Determine whether there are any outliers in the data. (2 marks)

c. Draw a box plot for this data. (3 marks)

3.3 Cumulative frequency

Practice Book, printed page 19.

1. The table shows the heights, in cm, of 70 teenagers.

Height, h (cm)	$130 \leq h < 140$	$140 \leq h < 150$	$150 \leq h < 160$	$160 \leq h < 170$
Frequency	15	18	23	14

a. Draw a cumulative frequency table for this data.

Hint

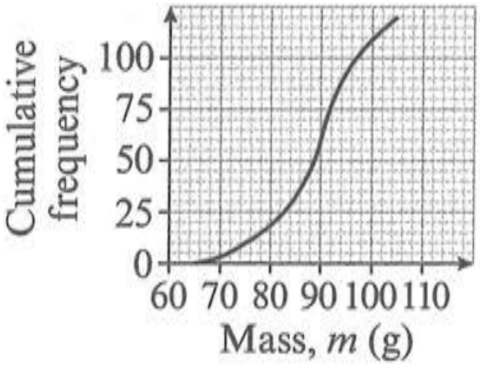
Add up the frequencies as you go along.

b. Draw a cumulative frequency diagram for this data.

Hint

Each point is plotted at the upper class boundary.

2. The cumulative frequency diagram shows the masses, in grams, of 120 apples.



Find an estimate for:

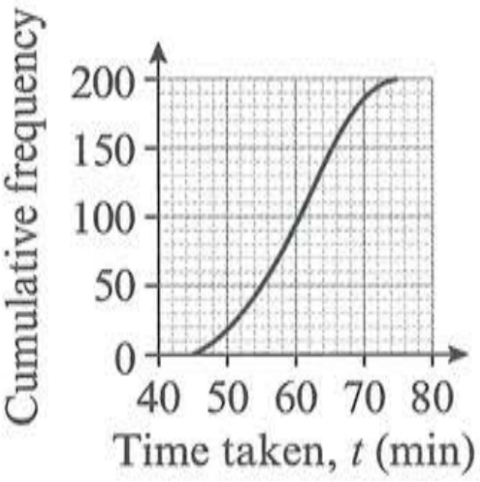
a. the median mass

b. the interquartile range.

Hint

Draw horizontal lines across to the curve from the cumulative frequency axis.

3. E The cumulative frequency diagram shows the times taken, in minutes, by a number of fun runners to complete an obstacle course.



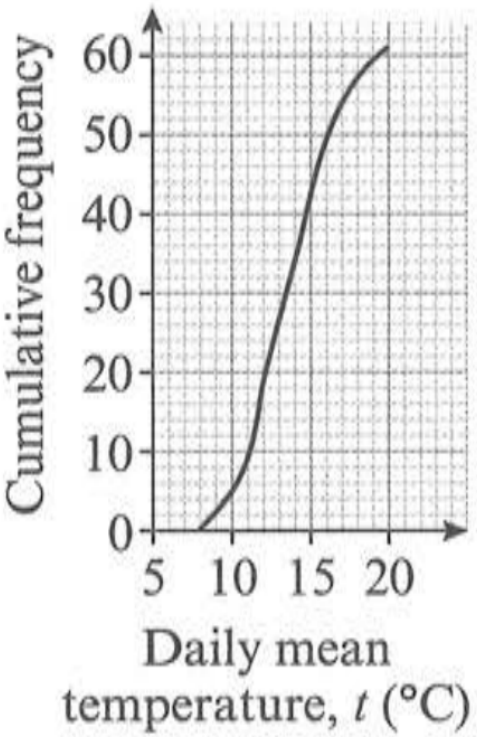
a. State how many runners took part. (1 mark)

b. Estimate the median time taken. (2 marks)

The qualifying time for the regional final was 58 minutes.

c. Estimate the number of runners who qualified for the regional final. (2 marks)

4. E/P From the large data set, the daily mean temperature (°C) in Hurn for the months of May and June 2015 is summarised in the cumulative frequency diagram.



a. Write down the number of days represented in this cumulative frequency diagram (1 mark)

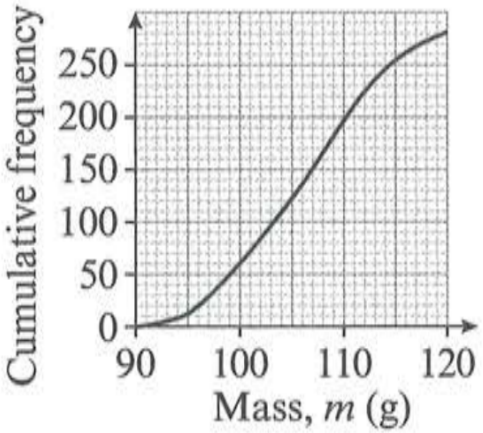
b. Estimate the median and interquartile range of the daily mean temperatures. (3 marks)

c. Estimate the number of days during this period that the temperature was greater than 15°C. (2 marks)

Phyllis says that the range of temperatures for the given period is 12°C.

d. Explain why Phyllis might be wrong. (2 marks)

5. E/P The masses, in grams, of a crop of potatoes are shown in the cumulative frequency diagram.



Only potatoes weighing between 102 g and 117 g can be sold to a supermarket.
Estimate the percentage of the crop that can be sold to the supermarket. (4 marks)

3.4 Histograms

Practice Book, printed page 20.

1. The table records the times taken by a group of Sudoku enthusiasts to complete a puzzle.

Time, t (min)	$3 \leq t < 4$	$4 \leq t < 4.5$	$4.5 \leq t < 5$	$5 \leq t < 6.5$	$6.5 \leq t < 8$
Frequency	7	13	19	15	6

a. Write down the class width of the $4 \leq t < 4.5$ interval.

b. Calculate the frequency densities for each class.

Hint

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

2. The table shows the length of time, t minutes, that it takes 41 cats to eat their dinner.

Time, t (min)	$0 \leq t < 3$	$3 \leq t < 4$	$4 \leq t < 4.5$	$4.5 \leq t < 5$	$5 \leq t < 6$	$6 \leq t < 10$
Frequency	3	6	8	7	5	12

a. Explain why a histogram is a suitable diagram for displaying this data.

b. Draw a histogram to represent the data.

Hint

Calculate the frequency densities first.

3. E/P The table shows the lengths of time, to the nearest ms, that it takes a group of students to respond to a stimulus.

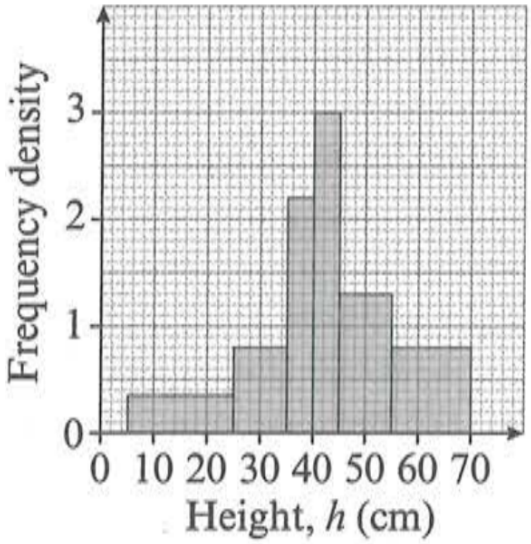
Time, t (ms)	$0 \leq t < 4$	$4 \leq t < 6$	$6 \leq t < 7$	$7 \leq t < 7.5$	$7.5 \leq t < 8.5$	$8.5 \leq t < 10$
Frequency	2	10	15	12	8	3

a. Give two reasons why a histogram is appropriate for displaying this data. (2 marks)

The bar to represent the students who took between 4 ms and 6 ms to respond is of width 1 cm and height 2.5 cm.

b. Calculate the width and height of the bar representing students who took between 7.5 ms and 8.5 ms to respond. (4 marks)

4. E/P The histogram shows the distribution of the heights of 66 garden gnomes in a garden centre.



a. Write down the number of gnomes that are between 25 cm and 35 cm tall. (2 marks)

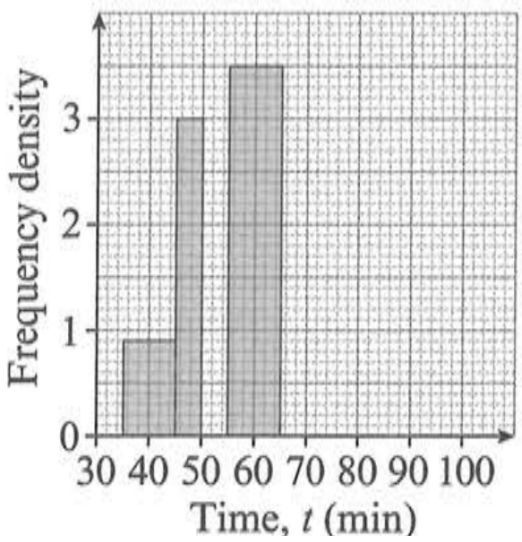
A gnome collector wants to buy all of the gnomes that are greater than 45 cm tall.

b. Calculate the percentage of the 66 gnomes that the collector will buy. (3 marks)

5. E/P The table shows the times taken, in minutes, by runners to complete a cross-country race.

Time, t (min)	$35 \leq t < 45$	$45 \leq t < 50$	$50 \leq t < 55$	$55 \leq t < 65$	$65 \leq t < 100$
Frequency	18	p	42	q	14

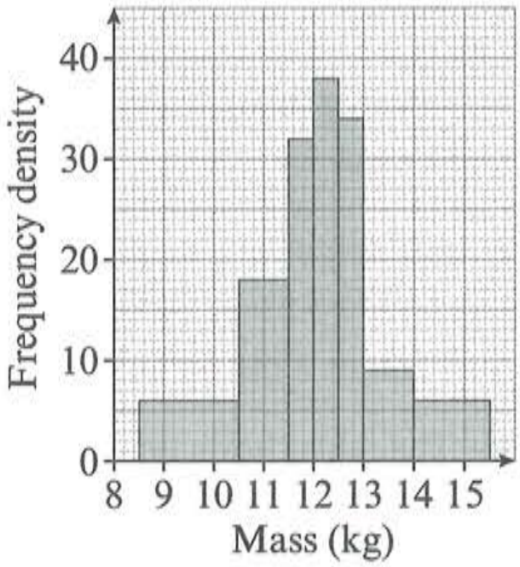
The partially completed histogram shows the same data.



a. Calculate the values of p and q. (3 marks)

b. Calculate the height of the bar representing runners who took between 50 and 55 minutes to complete the race. (2 marks)

6. E/P The histogram shows the masses, in kg, of 200 Malabar large-spotted civets.



Use interpolation to estimate the number of civets that are heavier than 10 kg. (4 marks)

3.5 Comparing data

Practice Book, printed page 22.

1. Summary statistics are calculated for the daily total hours of sunshine in a random sample taken from each of Hurn and Leeming during 1987. The results are shown in the table.

	Median	**Interquartile range**
Hurn	4.1 hours	5.6 hours
Leeming	4.9 hours	4.2 hours

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Write down two comparisons between the data from the two locations.

Hint

Compare a measure of location and a measure of spread.
Give your answers in the context of the data.

2. The times taken, in minutes, to complete a Kakuro puzzle are recorded for a random sample of year 8 students and a random sample of year 10 students. Summary statistics for each of the two data sets are recorded in the table.

	n	$\sum x$	$\sum x^2$
Year 8	15	94	610
Year 10	10	79	688

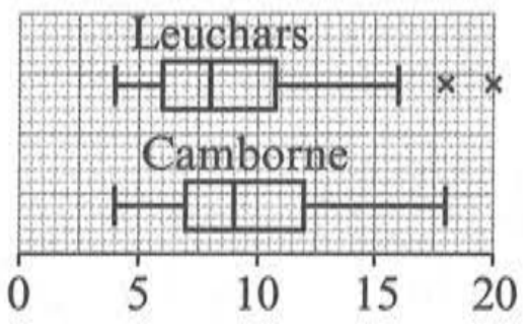
a. Calculate the mean and standard deviation of each data set.

b. Compare the two data sets.

Hint

Make sure you give two comparisons.

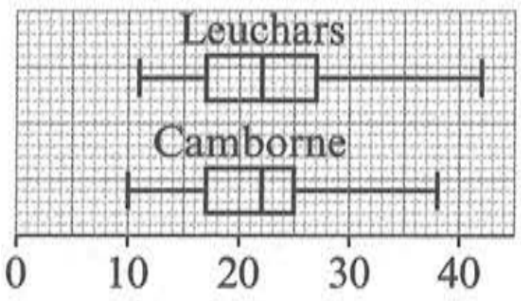
3. E/P The box plots show the distributions of daily mean windspeeds in Camborne and Leuchars during July and August 2015.



Anne and Colin want to go on a sailing holiday. To get the best out of the holiday, they want the windspeed to be as consistent as possible.

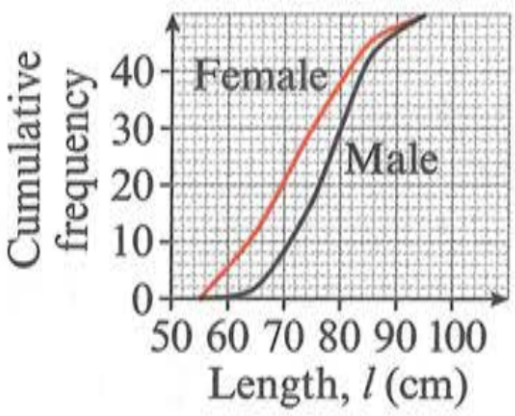
a. Suggest, with reasons, which of the two locations they should go to for their holiday. (2 marks)

These box plots show the data on daily maximum gusts in Camborne and Leuchars for the same two months.



b. Does this information change your recommendation? Give a reason for your answer. (2 marks)

4. E The cumulative frequency diagram shows the distribution of lengths in cm of 50 male and 50 female Eurasian otters.



Compare the lengths of male and female otters. (4 marks)

5. E/P A rodent expert records the masses, in grams, of colonies of brown and black rats. Her results are shown in the table.

Mass (g)	Frequency (black rats)	Frequency (brown rats)
$100 \leq m < 150$	2	1
$150 \leq m < 200$	7	8
$200 \leq m < 250$	18	32
$250 \leq m < 300$	24	22
$300 \leq m < 350$	13	7
$350 \leq m < 400$	6	0

a. Draw cumulative frequency diagrams for both sets of data on the same axes. (3 marks)

b. State which type of rat has the greater spread of masses. (2 marks)

A black rat is considered overweight if its mass is greater than 320 g and a brown rat is considered overweight if its mass is greater than 280 g.

c. Which colony contains the greatest number of overweight rats? (2 marks)

Problem solving: Set A

Bronze

The times taken, t seconds, for each of ten dogs to eat a biscuit are recorded.
The following summary statistics are calculated:

$$\sum t = 42 \qquad \sum t^2 = 216.4$$

An outlier is defined as being a data point that is greater than ± 2 standard deviations from the mean.
One of the dogs took 8.3 seconds to eat a biscuit.

a. Show that this piece of data is an outlier. (3 marks)

A student decides to remove this time from the data set before analysing the data further.

b. Suggest one reason why this may not be a valid approach. (1 mark)

Silver

A supermarket is testing the masses, m grams, of grapefruit. The masses of a random sample of 20 grapefruits are recorded, and the data is summarised as follows:

$$n = 20 \qquad \sum m = 3218 \qquad \sum m^2 = 530\,276$$

An outlier is defined as being a data point that is greater than ± 2 standard deviations from the mean.
The two largest masses recorded in the sample are 223 g and 845 g.
Benedict suggests that both of these values should be removed from the data set.
Use the definition of an outlier given above to evaluate Benedict's suggestion. (4 marks)

Gold

A chemical reaction is carried out 9 times. Each time the volume of reactant, in ml, is recorded. The results are shown below:

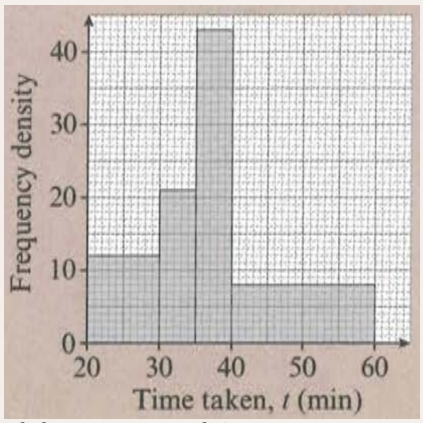
13 43 25 29 51 36 75 52 27

A researcher wants to identify extreme data values before carrying out her analysis. To do this, she proposes two different definitions of an outlier.
Definition A: a data point that is greater than ± 2 standard deviations from the mean.
Definition B: a data point that is more than $1.5\times$ the interquartile range below the lower quartile, or is more than $1.5\times$ the interquartile range above the upper quartile.
State, with reasons, which of these definitions would be most useful to the researcher. (5 marks)

Problem solving: Set B

Bronze

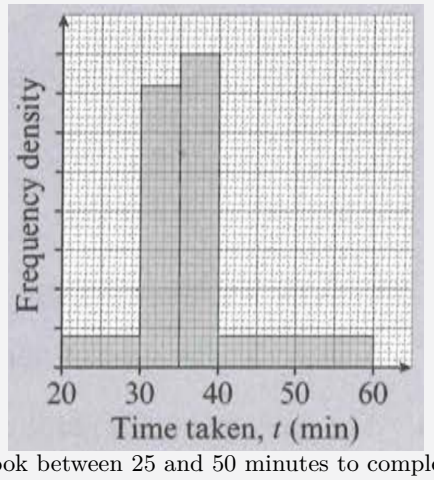
The histogram shows the times taken, in minutes, for a group of 600 students to complete a jigsaw puzzle.



Estimate the number of students who took between 32 and 42 minutes to complete the puzzle. (4 marks)

Silver

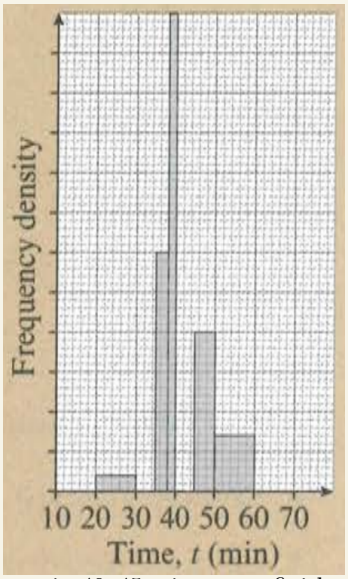
The histogram shows the times taken, in minutes, for a group of 100 runners to complete a cross-country race.



Estimate the number of runners who took between 25 and 50 minutes to complete the race. (4 marks)

Gold

A group of 120 swimmers completed an open water race and recorded their finishing times, t minutes. The times for two thirds of the participants are shown on the histogram. The bars for $30 \leq t < 35$ and $40 \leq t < 45$ are missing.



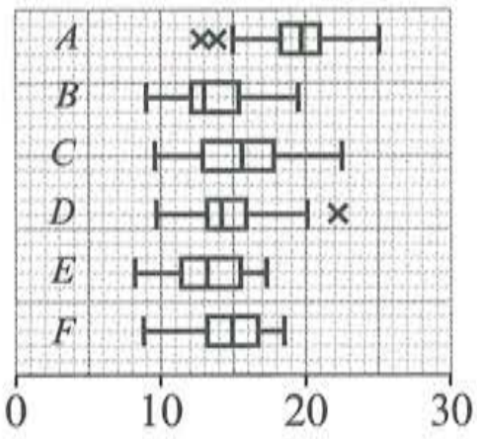
Four times as many swimmers finished the race in 40–45 minutes as finished the race in 30–35 minutes.
Estimate the number of swimmers who took between 25 and 43 minutes to complete the race. (6 marks)

Now try this → Exam question bank Q11, Q20, Q22, Q26, Q36

Selected exam question bank questions

Question 11

The box plots show the distributions of monthly daily mean temperatures, in °C, for Perth in 2015. The box plots are not in correct month order.



a. Using your knowledge of the large data set, suggest which two box plots represent the distribution of daily mean temperatures for July and August. (2 marks)

b. Explain what is meant by × in the box plots for months A and D. (1 mark)

c. Estimate the number of days in month C that had a daily mean temperature greater than 20°C. (2 marks)

Question 20

Subitsaa took a random sample of 30 daily mean windspeeds, in knots, from the large data set for Leeming in 2015. Her sample was stratified by month.

a. State how many data values Subitsaa sampled from the data for July 2015. (2 marks)

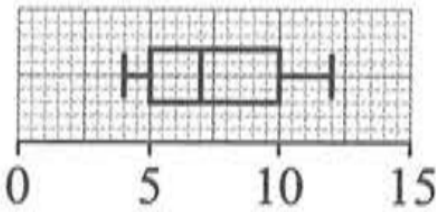
The data that Subitsaa collected is summarised in the table.

Windspeed, w (knots)	$4 \leq w < 6$	$6 \leq w < 8$	$8 \leq w < 10$	$10 \leq w < 12$
Frequency	8	9	6	7

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b. Use linear interpolation to estimate the median and interquartile range in Subitsaa's sample of daily mean windspeeds. (2 marks)

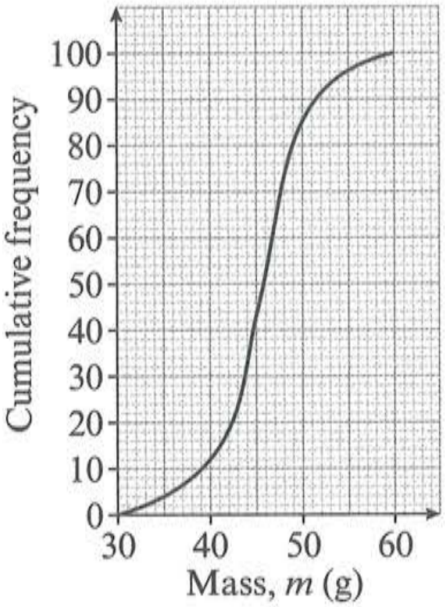
She took another random sample of daily mean windspeeds from the large data set for 30 days in Hurn in 2015. Her data is summarised in the box plot.



c. Compare the distribution of daily mean windspeeds, in knots, for Leeming and Hurn. (2 marks)

Question 22

The cumulative frequency diagram shows the distribution of masses, in grams, of 100 loggerhead shrikes.

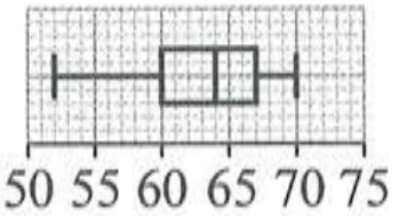


a. Estimate the median and interquartile range for this data. (3 marks)

A team of ornithologists wants to catch, ring and release any of the birds that are heavier than 51 g.

b. Estimate the number of loggerhead shrikes that they need to catch. (2 marks)

The box plot shows the distribution of masses, in grams, of 100 great grey shrikes.

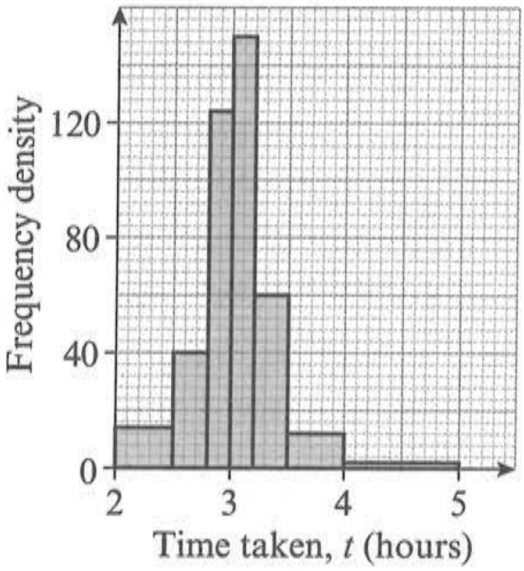


The ornithologists want to catch, ring and release the same number of great grey shrikes as loggerhead shrikes. They estimate that the great grey shrikes have to have a mass greater than x g.

c. Use interpolation to calculate the value of x. (2 marks)

Question 26

The histogram shows the distribution of the times taken, in hours, for 1000 athletes to complete a marathon.



The runners qualify for the national championships if they finish the marathon in less than 2 hours and 36 minutes.

a. Estimate the number of runners who qualified for the championships. (3 marks)

b. Estimate the mean and standard deviation of the times taken. (4 marks)

Question 36

Tanvi is researching the distances travelled by people to her local church. She stands outside the door of the church on Sunday between 10 am and 10:30 am and asks the first 20 people, as they arrive, how far they have travelled.

a. State the sampling method Tanvi has used. (1 mark)

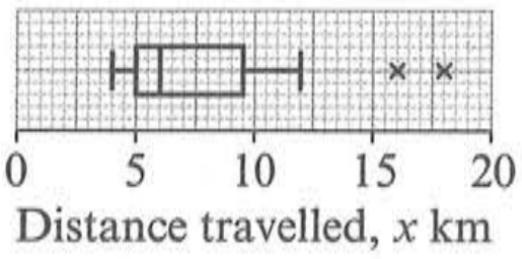
b. State and briefly describe an alternative non-random sampling method that Tanvi could have used to collect her sample of 20 people. (2 marks)

Spardha decides to ask every person that attends church on Sunday the distance, x km, that they have travelled.

c. State the data selection process that Spardha has used. (1 mark)

Spardha's results are summarised by the box plot and summary statistics below.

$$n = 50, \quad \sum x = 385, \quad \sum x^2 = 3517$$



d. Write down the interquartile range for these data. (1 mark)

e. Calculate the mean and standard deviation for these data. (3 marks)

f. State, giving a reason, whether you would recommend using the mean and standard deviation, or the median and interquartile range, to describe these data. (2 marks)

Joan and Esha both attend the church and have moved house since Spardha collected her data. Joan's journey has changed from a distance of 4.5 km to 9 km and Esha's journey has changed from a distance of 4.8 km to 8 km.

Spardha redrew her box plot and only had to change two values.

g. Explain which two values Spardha had to change and whether they had increased or decreased. (3 marks)

Answers

3.1 Outliers

1.

a. Yes
b. No
c. Yes

2.

a. No
b. Yes
c. No

3.

a. Yes; greater than 2 standard deviations from the mean.
b. 1.9 m; 1.1 m
c. No; it could be a legitimate data value, for example, a very mature adult snake.

4.

a. 0; 82
b. There are people that live to be 103 years old.

5.

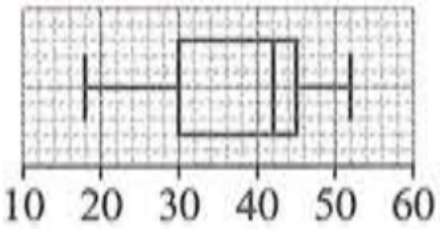
a. 15.4; 2.58 (3 s.f.)
b. Yes; greater than 20.56.
c. 21 is too old to be a secondary school student.
d. 14.8; 1.87 (3 s.f.)

6.

a. 28.9°C; 5.15°C (3 s.f.)
b. Outliers are less than 18.6 and greater than 39.2, so 18.5 and 42.6 are outliers.
c. 18.5 no, it could just be a very cold day; 42.6 yes, it is an extreme value that does not fit the data.
d. Mean 27.6°C, standard deviation 3.21°C (3 s.f.)

3.2 Box plots

1.



2.

a. 51 goals

b. 28 goals

c. 57 goals

3.

a. An outlier

b. Spread of masses is very similar but females, on average, weigh more than males.

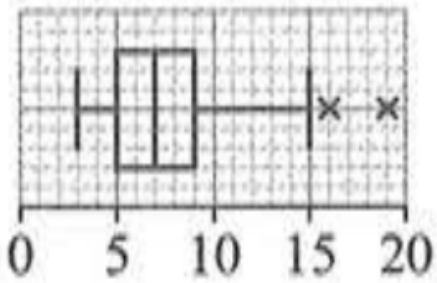
c. Female: only one male bird has a mass greater than this, but between 25% and 50% of females have a mass greater than 3.6 g.

4.

a. Median = 7 knots, interquartile range = 4 knots

b. 16 and 19 are outliers.

c.

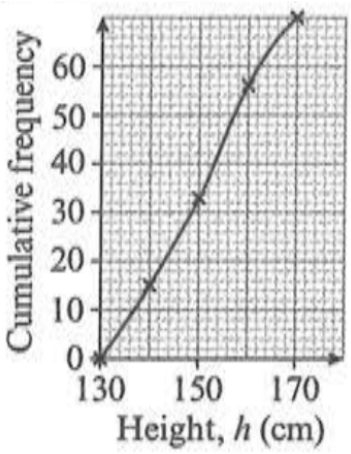


3.3 Cumulative frequency

1.

a.

Height (cm)	Cumulative frequency
$130 \leq h < 140$	15
$140 \leq h < 150$	33
$150 \leq h < 160$	56
$160 \leq h < 170$	70



2.

a. 90 g

b. 10.4 g

3.

a. 200

b. 60.5 minutes

c. 76 runners

4.

a. 61 days

b. Median 13.5°C, interquartile range 4.1°C

c. 19 days

d. She is assuming that the lowest recorded temperature is 8°C and the highest recorded temperature is 20°C but this might not be the case since the data is grouped. The lowest temperature could actually be 10°C and the highest any value greater than 18°C.

5. 66% (approximately)

3.4 Histograms

1.

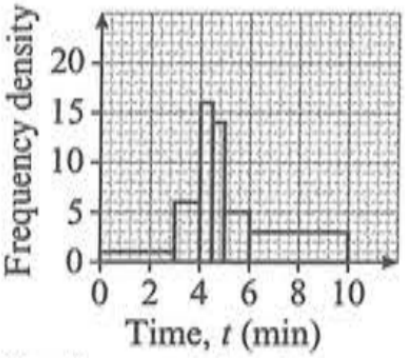
a. 0.5

b. 7, 26, 38, 10, 4

2.

a. The classes are of unequal width.

b.



3.

a. Continuous data; classes are of different widths.

b. Width 0.5 cm, height 4 cm

4.

a. 8

b. 37.9% (3 s.f.)

5.

a. $p = 30$, $q = 70$

b. 4.2

6. 182

3.5 Comparing data

1. The daily total sunshine in Hurn is on average lower but has greater variability.

2.

a. Year 8: mean 6.3 minutes (2 s.f.), standard deviation 1.2 minutes (2 s.f.)
Year 10: mean 7.9 minutes, standard deviation 2.5 minutes (2 s.f.)

b. Year 10 has a higher average time and the times are more variable.

3.

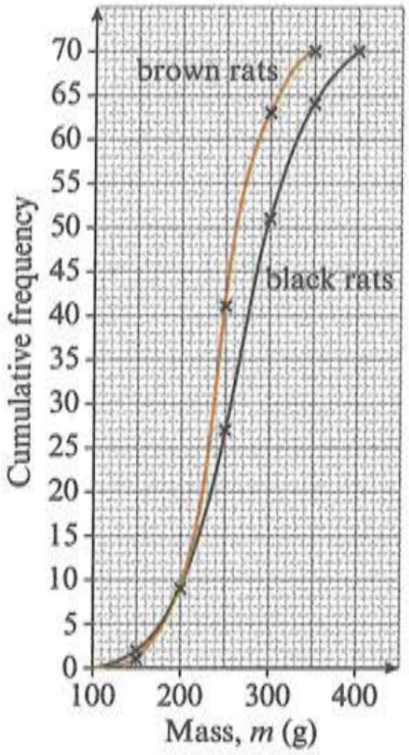
a. Leuchars: slightly smaller interquartile range and smaller range if the two outliers are ignored.

b. Yes: less variability in the gustiness of the wind for Camborne.

4. Male otters are generally longer (median approximately 78 compared to 72) and females have a greater variability in length (interquartile range approximately 15 compared to 11).

5.

a.



b. Black rats

c. Brown rats

Problem solving: Set A

Bronze

a. Mean 4.2 seconds, standard deviation 2 seconds
$4.2 + 2 \times 2 = 8.2$, and $8.3 > 8.2$, so is an outlier.

b. This is a reasonable amount of time taken to eat the biscuit, so it could be a legitimate data value.

Silver

Mean 160.9 g, standard deviation 25.0 g (3 s.f.)

Both values are outliers, but 223 g is a reasonable mass for a grapefruit, while 845 g is a much more extreme value so is likely to be an error. Only remove the 845 g value.

Gold

Mean 39 ml, standard deviation 17.5 ml (3 s.f.)

$Q_1 = 27$, $Q_3 = 51$, interquartile range 24

Definition A produces one outlier (75).

Definition B produces no outliers.

Definition A is a more useful method of identifying extreme data values because it sets a lower and upper bound for acceptable data values. Definition B only sets an upper bound; $27 - 1.5 \times 24 = -9$, so even a value of 0 would not be regarded as an extreme value under this definition.

Problem solving: Set B

Bronze 294 students

Silver 88 runners

Gold 71 or 72 swimmers

Selected exam question bank answers

Question 11

a. B and E

b. Outliers

c. 4

Question 20

a. 5

b. Median 7.56 kn, interquartile range 3.96 kn (3 s.f.)

c. Hurn has lower daily mean windspeeds on average and a greater variability in daily mean windspeeds.

Question 22

a. Median 46 g, interquartile range 5 g

b. 11

c. 68.68

Question 26

a. 110 runners

b. Mean 3.03 hours, standard deviation 0.290 hours (3 s.f.)

Question 36

a. Opportunity sampling

b. Quota sampling. She could ask the first people who fulfil certain characteristics, such as gender and age, until she has reached her quota for each type.

c. Census

d. 4.5 km

e. Mean 7.7 km; standard deviation 3.32 km (3 s.f.)

f. Median and interquartile range; since the data is skewed the mean and standard deviation are affected by the extreme values.

g. Lower quartile and median will both increase since data values have moved from below the lower quartile to above the median.